Ex. No: 1 **Distributed Synchronization Controller**

PROGRAM:

```
import java.util.Scanner;
public class Main {
   public static int gcd(int a, int b) {
      while (b != 0) {
         int temp = b;
         b = a % b;
         a = temp;
      }
      return a;
   }
   public static void main(String[] args) {
      Scanner sc = new Scanner(System.in);
      int a = sc.nextInt();
      int b = sc.nextInt();
      int gcd = gcd(a, b);
      long lcm = (long) a * b / gcd;
      System.out.println("Greatest Synchronization Unit:" + gcd);
      System.out.println("Unified Execution Cycle:" + lcm);
      sc.close();
   }
}
```

Ex. No: 2 **Autonomous Identity Verification**

PROGRAM:

```
import java.util.Scanner;
public class Solution {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        long square = (long) n * n;
        int temp = n;
        int digits = 1;
        while (temp >= 10) {
            digits++;
            temp /= 10;
        }
        long divisor = 1;
        for (int i = 0; i < digits; i++) {
            divisor *= 10;
        }
        if (square % divisor == n) {
            System.out.println("Self-Consistent Identity");
        } else {
            System.out.println("Inconsistent Identity");
        }
    }
}
```

Ex. No: 3 **Intelligent Resource Allocation Validator**

PROGRAM:

```
import java.util.Scanner;
public class Solution {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        int temp = n;
        int sum = 0;
        while (temp > 0) {
            sum += temp % 10;
            temp /= 10;
        }
        if (n % sum == 0) {
            System.out.println("Balanced Allocation");
        } else {
            System.out.println("Unbalanced Allocation");
        }
    }
}
```

Ex. No: 4 **Binary Transmission Encoder**

PROGRAM:

```
import java.util.Scanner;
public class Solution {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        if (n == 0) {
            System.out.println(0);
            return;
        }
        String binary = " ";
        while (n > 0) {
            binary = (n % 2) + binary;
            n /= 2;
        }
        System.out.println(binary);
    }
}
```

Ex. No: 5 **Transmission Priority Analyzer**

PROGRAM:

```java
import java.util.Scanner;
public class Solution {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        if (n == 0) {
            System.out.println(0);
            return;
        }
        int maxDigit = 0;
        while (n > 0) {
            int digit = n % 10;
            if (digit > maxDigit) {
                maxDigit = digit;
            }
            n /= 10;
        }

        System.out.println(maxDigit);
    }
}
```

Ex. No: 6 **Rectangular Simulation Matrix**

PROGRAM:

```
import java.util.Scanner;
public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int rows = sc.nextInt();
        int cols = sc.nextInt();
        int num = 2;
        for (int i = 0; i < rows; i++) {
            for (int j = 0; j < cols; j++) {
                System.out.print(num);
                if (j < cols - 1) {
                    System.out.print(" ");
                }
                num += 2;
            }
            System.out.println();
        }
        sc.close();
    }
}
```

Ex. No: 7 **Cone Multiplication Pattern**

PROGRAM:

```
import java.util.Scanner;
public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        int num = 3;
        for (int i = 1; i <= n; i++) {
            for (int j = 1; j <= (n - i) * 3; j++) {
                System.out.print(" ");
            }
            for (int j = 1; j <= (2 * i - 1); j++) {
                System.out.print(num);
                num += 3;
                if (j < (2 * i - 1)) {
                    System.out.print(" ");
                }
            }

            System.out.println();
        }
        sc.close();
    }
}
```

Ex. No: 8 **Checkboard Pattern 5**

PROGRAM:

```java
import java.util.Scanner;
public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        for (int i = 0; i < n; i++) {
            for (int j = 0; j < n; j++) {
                if ((i + j) % 2 == 0) {
                    System.out.print("*");
                } else {
                    System.out.print("#");
                }
                if (j < n - 1) {
                    System.out.print(" ");
                }
            }
            System.out.println();
        }
        sc.close();
    }
}
```

Ex. No: 9 **Hollow Number Square Pattern**

PROGRAM:

```
import java.util.Scanner;
public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        int num = 1;
        for (int i = 0; i < n; i++) {
            for (int j = 0; j < n; j++) {
                if (i == 0 || i == n - 1 || j == 0 || j == n - 1) {
                    System.out.print(num);
                    num++;
                } else {
                    System.out.print(" ");
                }
                if (j != n - 1) {
                    System.out.print(" ");
                }
            }
            System.out.println();
        }
        sc.close();
    }
}
```

Ex. No: 10 **ZigZag Pattern 10**

PROGRAM:

```
import java.util.Scanner;
public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int rows = sc.nextInt();
        int cols = sc.nextInt();
        int num = 1;
        for (int i = 0; i < rows; i++) {
            int[] arr = new int[cols];
            for (int j = 0; j < cols; j++) {
                arr[j] = num++;
            }
            if (i % 2 == 0) {
                for (int j = 0; j < cols; j++) {
                    System.out.print(arr[j]);
                    if (j < cols - 1)
                        System.out.print(" ");
                }
            }
            else {
                for (int j = cols - 1; j >= 0; j--) {
                    System.out.print(arr[j]);
                    if (j > 0)
                        System.out.print(" ");
```

```
                }
            }

            System.out.println();
        }
        sc.close();
    }
}
```